

Introduction to Artificial Intelligence and Machine Learning

What is Artificial Intelligence? Artificial Intelligence (AI) is a field of computer science focused on building systems that can perform tasks that normally require human intelligence such as learning, reasoning, problem-solving, understanding language, and perception.

What is Machine Learning? Machine Learning (ML) is a subset of AI that enables systems to learn from data instead of being explicitly programmed. The three main types are Supervised Learning, Unsupervised Learning, and Reinforcement Learning.

Key Concepts in Machine Learning include Features, Labels, Training Data, and Testing Data. Features are inputs used for prediction, while labels are the outputs a model learns to predict.

Natural Language Processing (NLP) is a branch of AI that allows computers to understand and process human language. Applications include chatbots, sentiment analysis, translation, and summarization.

Large Language Models (LLMs) are advanced AI systems trained on massive text data to generate human-like language. They can answer questions, write essays, summarize documents, and generate code.

The Problem of Hallucination in AI refers to when LLMs produce incorrect or fabricated information. This happens because models predict likely text rather than verified facts.

Retrieval-Augmented Generation (RAG) improves LLM accuracy by retrieving relevant information from external documents before generating answers.

Embeddings convert text into numerical vectors representing meaning. Vector databases store these embeddings so AI systems can search based on meaning rather than exact keywords.

Applications of RAG Systems include company knowledge assistants, legal document search, medical record analysis, and customer support bots.

Ethical Considerations in AI include bias, privacy, misuse prevention, and transparency. AI should assist humans and be used responsibly.